# संभव BATCH

## Swachhand IAS Academy | स्वछंद IAS

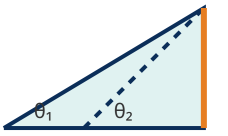
*Under the guidance of Vipulendra Tripathi Sir*

## MATHEMATICS: HEIGHT & DISTANCE

### Question 1

SSC CGL Tier-II 2022

From the top of a 150 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




**(A) 150 m**

**(B) 150(√3 + 1) m**

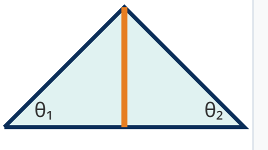
**(C) 150√3 m**

**(D) 150(√3 - 1) m**

### Question 2

SSC CGL Tier-II 2020

Two ships are navigating in the sea on opposite sides of a lighthouse of height 120 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 45° respectively. Find the distance between the two ships.




**(A) 120(√3 + 1) m**

**(B) 120(√3 - 1) m**

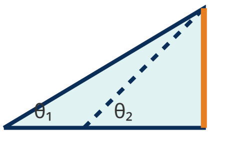
**(C) 120 m**

**(D) 120√3 m**

### Question 3

SSC CGL Tier-II 2020

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 45°. If the height of the tower is 30 m, find the difference in the lengths of the shadow.




**(A) 30(√3 - 1) m**

**(B) 30(√3 + 1) m**

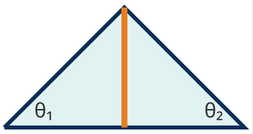
**(C) 30 m**

**(D) 30√3 m**

## Question 4

SSC CGL Tier-I 2019

Two ships are navigating in the sea on opposite sides of a lighthouse of height 100 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 45° respectively. Find the distance between the two ships.




**(A) $100(\sqrt{3} + 1)$ m**

**(B) $100\sqrt{3}$ m**

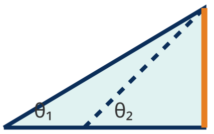
**(C) $100(\sqrt{3} - 1)$ m**

**(D) 100 m**

## Question 5

SSC CGL Tier-II 2022

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 45°. If the height of the tower is 20 m, find the difference in the lengths of the shadow.




**(A) $20\sqrt{3}$ m**

**(B) $20(\sqrt{3} + 1)$ m**

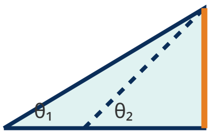
**(C) 20 m**

**(D) $20(\sqrt{3} - 1)$ m**

## Question 6

SSC CGL Tier-II 2020

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 60°. If the height of the tower is 10 m, find the difference in the lengths of the shadow.




**(A) 10/2 m**

**(B) $10(\sqrt{3})$ m**

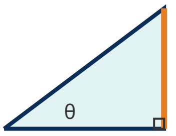
**(C) $10(2/\sqrt{3})$ m**

**(D) 20 m**

## Question 7

SSC CGL Tier-I 2021

The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 60°. Find the height of the tower.




**(A) 30 m**

**(B) $30\sqrt{3}$ m**

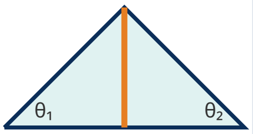
**(C) $60/\sqrt{3}$ m**

**(D) $30/\sqrt{3}$ m**

## Question 8

SSC CGL Tier-II 2020

Two ships are navigating in the sea on opposite sides of a lighthouse of height 100 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 60° respectively. Find the distance between the two ships.



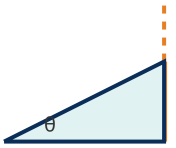

**(A) 100(2/√3) m**  **(B) 100/√3 m**

**(C) 100(4/√3) m**  **(D) 400 m**

## Question 9

SSC CGL Tier-I 2021

A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 10 m. Find the original height of the tree.



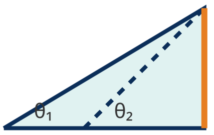

**(A) 30/√3 m**  **(B) 20 m**

**(C) 10/√3 m**  **(D) 10√3 m**

## Question 10

SSC CGL Tier-I 2022

From the top of a 200 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.



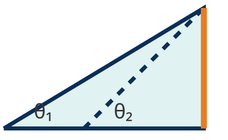

**(A) 200√3 m**  **(B) 200 m**

**(C) 200(√3 - 1) m**  **(D) 200(√3 + 1) m**

## Question 11

SSC CGL Tier-II 2021

The angles of elevation of the top of a tower from two points at a distance of 16 m and 25 m from the base of the tower and in the same straight line with it are complementary. What is the height of the tower?




(A) 41 m

(B) 9 m

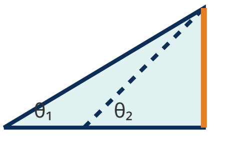
(C) 20 m

(D) 6 m

## Question 12

SSC CGL Tier-I 2019

The angles of elevation of the top of a tower from two points at a distance of 16 m and 25 m from the base of the tower and in the same straight line with it are complementary. What is the height of the tower?




(A) 6 m

(B) 20 m

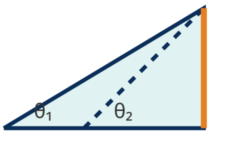
(C) 9 m

(D) 41 m

## Question 13

SSC CGL Tier-I 2019

From the top of a 150 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




(A) $150\sqrt{3}$ m

(B) $150(\sqrt{3} + 1)$ m

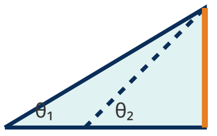
(C) 150 m

(D) $150(\sqrt{3} - 1)$ m

### Question 14

SSC CGL Tier-I 2023

The angles of elevation of the top of a tower from two points at a distance of 4 m and 49 m from the base of the tower and in the same straight line with it are complementary. What is the height of the tower?




(A) 53 m

(B) 7 m

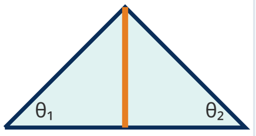
(C) 45 m

(D) 14 m

### Question 15

SSC CGL Tier-I 2019

Two ships are navigating in the sea on opposite sides of a lighthouse of height 100 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 45° respectively. Find the distance between the two ships.




(A) 100 m

(B) $100(\sqrt{3} - 1)$ m

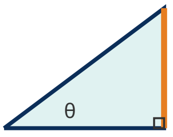
(C) $100(\sqrt{3} + 1)$ m

(D) $100\sqrt{3}$ m

### Question 16

SSC CGL Tier-II 2022

The angle of elevation of the top of a tower from a point on the ground, which is 100 m away from the foot of the tower, is 45°. Find the height of the tower.




(A) $100/\sqrt{3}$ m

(B) $100\sqrt{3}$ m

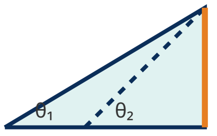
(C) 100 m

(D) 50.0 m

### Question 17

SSC CGL Tier-I 2021

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 45° than when it is 60°. If the height of the tower is 40 m, find the difference in the lengths of the shadow.




(A) $40(1 - 1/\sqrt{3})$ m

(B) $40/\sqrt{3}$ m

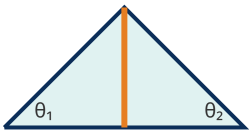
(C) 40 m

(D) $40(1 + 1/\sqrt{3})$ m

## Question 18

SSC CGL Tier-II 2022

Two ships are navigating in the sea on opposite sides of a lighthouse of height 80 m. The angles of depression of the two ships from the top of the lighthouse are 45° and 60° respectively. Find the distance between the two ships.




(A) 80 m

(B) 80/√3 m

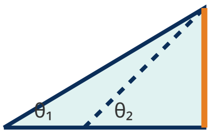
(C) 80(1 + 1/√3) m

(D) 80(1 - 1/√3) m

## Question 19

SSC CGL Tier-I 2019

The angles of elevation of the top of a tower from two points at a distance of 16 m and 36 m from the base of the tower and in the same straight line with it are complementary. What is the height of the tower?




(A) 7 m

(B) 20 m

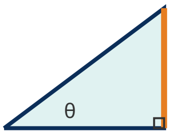
(C) 52 m

(D) 24 m

## Question 20

SSC CGL Tier-II 2022

The angle of elevation of the top of a tower from a point on the ground, which is 60 m away from the foot of the tower, is 60°. Find the height of the tower.




(A) 120/√3 m

(B) 60 m

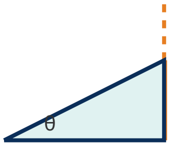
(C) 60/√3 m

(D) 60√3 m

## Question 21

SSC CGL Tier-II 2022

A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 60° with it. The distance between the foot of the tree to the point where the top touches the ground is 10 m. Find the original height of the tree.




(A) 10√3 m

(B) 10 m

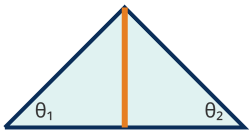
(C) 10(2 + √3) m

(D) 10(2 - √3) m

## Question 22

SSC CGL Tier-II 2021

Two ships are navigating in the sea on opposite sides of a lighthouse of height 100 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 60° respectively. Find the distance between the two ships.




(A) $100(4/\sqrt{3})$ m

(B) 400 m

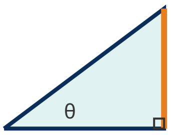
(C) $100(2/\sqrt{3})$ m

(D) $100/\sqrt{3}$ m

## Question 23

SSC CGL Tier-I 2022

The angle of elevation of the top of a tower from a point on the ground, which is 75 m away from the foot of the tower, is 30°. Find the height of the tower.




(A) $75\sqrt{3}$ m

(B) 150 m

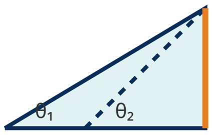
(C) 75 m

(D) $75/\sqrt{3}$ m

## Question 24

SSC CGL Tier-I 2021

From the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




(A) $100(\sqrt{3} - 1)$ m

(B) 100 m

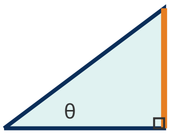
(C) $100(\sqrt{3} + 1)$ m

(D) $100\sqrt{3}$ m

## Question 25

SSC CGL Tier-II 2020

The angle of elevation of the top of a tower from a point on the ground, which is 150 m away from the foot of the tower, is 30°. Find the height of the tower.




(A) 150 m

(B) $150\sqrt{3}$ m

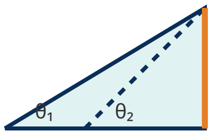
(C) 300 m

(D) $150/\sqrt{3}$ m

## Question 26

SSC CGL Tier-II 2021

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 60°. If the height of the tower is 40 m, find the difference in the lengths of the shadow.




**(A) 40/2 m**

**(B) 80 m**

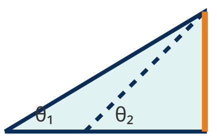
**(C) 40(2/√3) m**

**(D) 40(√3) m**

## Question 27

SSC CGL Tier-I 2023

From the top of a 150 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




**(A) 150(√3 - 1) m**

**(B) 150 m**

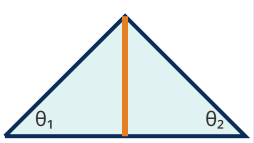
**(C) 150√3 m**

**(D) 150(√3 + 1) m**

## Question 28

SSC CGL Tier-I 2019

Two ships are navigating in the sea on opposite sides of a lighthouse of height 80 m. The angles of depression of the two ships from the top of the lighthouse are 30° and 45° respectively. Find the distance between the two ships.




**(A) 80(√3 + 1) m**

**(B) 80(√3 - 1) m**

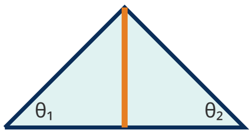
**(C) 80 m**

**(D) 80√3 m**

## Question 29

SSC CGL Tier-I 2019

Two ships are navigating in the sea on opposite sides of a lighthouse of height 300 m. The angles of depression of the two ships from the top of the lighthouse are 45° and 60° respectively. Find the distance between the two ships.




(A) $300/\sqrt{3}$ m

(B) $300(1 - 1/\sqrt{3})$ m

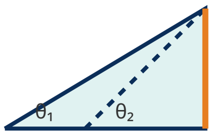
(C) $300(1 + 1/\sqrt{3})$ m

(D) 300 m

## Question 30

SSC CGL Tier-II 2020

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 60°. If the height of the tower is 20 m, find the difference in the lengths of the shadow.




(A) $20(2/\sqrt{3})$ m

(B) $20(\sqrt{3})$ m

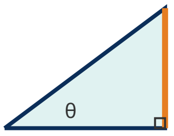
(C) 20/2 m

(D) 40 m

## Question 31

SSC CGL Tier-II 2020

The angle of elevation of the top of a tower from a point on the ground, which is 60 m away from the foot of the tower, is 45°. Find the height of the tower.




(A) $60\sqrt{3}$ m

(B) $60/\sqrt{3}$ m

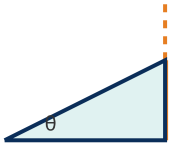
(C) 30.0 m

(D) 60 m

## Question 32

SSC CGL Tier-II 2020

A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 15 m. Find the original height of the tree.




(A) 30 m

(B) $15/\sqrt{3}$ m

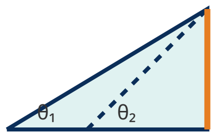
(C) $15\sqrt{3}$ m

(D) $45/\sqrt{3}$ m

## Question 33

SSC CGL Tier-II 2021

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 45° than when it is 60°. If the height of the tower is 40 m, find the difference in the lengths of the shadow.




**(A) 40(1 - 1/√3) m**

**(B) 40(1 + 1/√3) m**

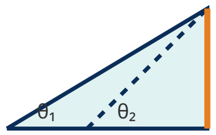
**(C) 40/√3 m**

**(D) 40 m**

## Question 34

SSC CGL Tier-II 2020

The angles of elevation of the top of a tower from two points at a distance of 4 m and 25 m from the base of the tower and in the same straight line with it are complementary. What is the height of the tower?




**(A) 5 m**

**(B) 10 m**

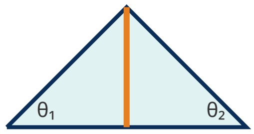
**(C) 29 m**

**(D) 21 m**

## Question 35

SSC CGL Tier-II 2021

Two ships are navigating in the sea on opposite sides of a lighthouse of height 300 m. The angles of depression of the two ships from the top of the lighthouse are 45° and 60° respectively. Find the distance between the two ships.




**(A) 300(1 - 1/√3) m**

**(B) 300 m**

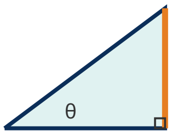
**(C) 300(1 + 1/√3) m**

**(D) 300/√3 m**

## Question 36

SSC CGL Tier-I 2023

The angle of elevation of the top of a tower from a point on the ground, which is 150 m away from the foot of the tower, is 60°. Find the height of the tower.




**(A) 300/√3 m**

**(B) 150 m**

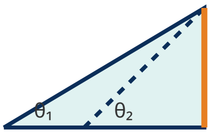
**(C) 150/√3 m**

**(D) 150√3 m**

## Question 37

SSC CGL Tier-II 2020

From the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




**(A) 200 m**

**(B) 100(√3) m**

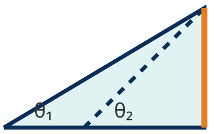
**(C) 100(2/√3) m**

**(D) 100/2 m**

## Question 38

SSC CGL Tier-I 2019

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 45°. If the height of the tower is 30 m, find the difference in the lengths of the shadow.




**(A) 30(√3 + 1) m**

**(B) 30√3 m**

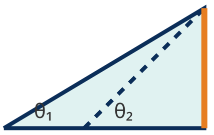
**(C) 30(√3 - 1) m**

**(D) 30 m**

## Question 39

SSC CGL Tier-I 2022

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 45° than when it is 60°. If the height of the tower is 40 m, find the difference in the lengths of the shadow.




**(A) 40 m**

**(B) 40/√3 m**

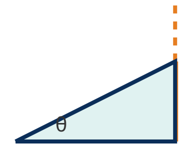
**(C) 40(1 + 1/√3) m**

**(D) 40(1 - 1/√3) m**

## Question 40

SSC CGL Tier-I 2023

A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 60° with it. The distance between the foot of the tree to the point where the top touches the ground is 10 m. Find the original height of the tree.




**(A) 10(2 - √3) m**

**(B) 10(2 + √3) m**

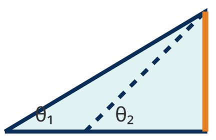
**(C) 10√3 m**

**(D) 10 m**

## Question 41

SSC CGL Tier-I 2019

From the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




**(A) 100(2/√3) m**

**(B) 200 m**

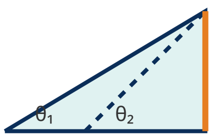
**(C) 100/2 m**

**(D) 100(√3) m**

## Question 42

SSC CGL Tier-I 2021

From the top of a 150 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.




**(A) 150(√3 - 1) m**

**(B) 150(√3 + 1) m**

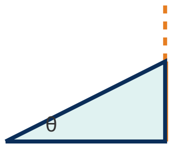
**(C) 150 m**

**(D) 150√3 m**

## Question 43

SSC CGL Tier-I 2019

A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 60° with it. The distance between the foot of the tree to the point where the top touches the ground is 30 m. Find the original height of the tree.




**(A) 30 m**

**(B) 30√3 m**

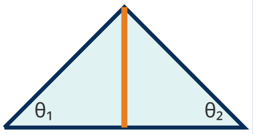
**(C) 30(2 - √3) m**

**(D) 30(2 + √3) m**

## Question 44

SSC CGL Tier-I 2019

Two ships are navigating in the sea on opposite sides of a lighthouse of height 80 m. The angles of depression of the two ships from the top of the lighthouse are 45° and 60° respectively. Find the distance between the two ships.




**(A) 80/√3 m**

**(B) 80(1 + 1/√3) m**

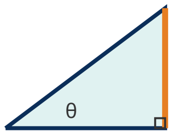
**(C) 80(1 - 1/√3) m**

**(D) 80 m**

## Question 45

SSC CGL Tier-II 2021

The angle of elevation of the top of a tower from a point on the ground, which is 60 m away from the foot of the tower, is 60°. Find the height of the tower.




**(A) 120/√3 m**

**(B) 60√3 m**

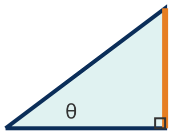
**(C) 60 m**

**(D) 60/√3 m**

## Question 46

SSC CGL Tier-II 2021

The angle of elevation of the top of a tower from a point on the ground, which is 60 m away from the foot of the tower, is 45°. Find the height of the tower.




**(A) 60/√3 m**

**(B) 60 m**

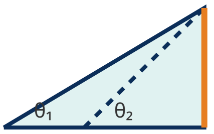
**(C) 30.0 m**

**(D) 60√3 m**

## Question 47

SSC CGL Tier-I 2023

From the top of a 150 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

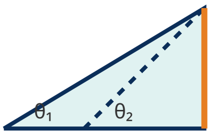



(A) 150(√3) m
(B) 300 m
(C) 150/2 m
(D) 150(2/√3) m

## Question 48

SSC CGL Tier-I 2023

From the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are 45° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

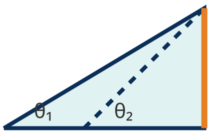



(A) 100(1 + 1/√3) m
(B) 100(1 - 1/√3) m
(C) 100 m
(D) 100/√3 m

## Question 49

SSC CGL Tier-I 2022

The shadow of a tower standing on a level ground is found to be certain meters longer when the sun's altitude is 30° than when it is 60°. If the height of the tower is 50 m, find the difference in the lengths of the shadow.

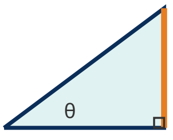



(A) 50/2 m
(B) 100 m
(C) 50(√3) m
(D) 50(2/√3) m

## Question 50

SSC CGL Tier-II 2021

The angle of elevation of the top of a tower from a point on the ground, which is 50 m away from the foot of the tower, is 30°. Find the height of the tower.




(A) 100 m
(B) 50 m
(C) 50/√3 m
(D) 50√3 m

**स्वच्छंद सोच, सर्वश्रेष्ठ परिणाम**